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Empirical Persona Validation of the Isocline Mobile Application:

A Programmatic Walkthrough Study of Seven Hormone Replacement Therapy Regimens Against Live Pharmacokinetic Model Output

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Abstract

We conduct an empirical validation of Isocline, a mobile application for hormone replacement therapy (HRT) monitoring, by implementing seven synthetic user personas as runnable test profiles within the application and examining the resulting interface state against the design claims of the companion UX paper [2]. Each persona is constructed with a complete medication configuration, dose history, blood test records, and journal entries, producing a full pharmacokinetic (PK) model prediction that can be interrogated programmatically. We report the home screen state—estimated hormone level, classification status, cycle position, and therapeutic comparison—for each persona and analyze the results against the design principles of progressive disclosure, honest presentation, and per-user personalization.

The study reveals both strengths and gaps. The presentation rules correctly gate comparative language behind user-set therapeutic targets: personas without configured targets (Sam, Rowan) receive neutral level estimates without "below target" judgments, while personas with targets receive appropriate comparisons. Regimen changes (Priya's route switch, Devon's interval doubling) are handled without data loss, and the pharmacokinetic output reproduces the companion model paper's predictions in every case we recomputed. The transdermal personas (Sam, Rowan) show `unknown` cycle position, indicating that the phase classifier does not handle daily-application routes.

Two findings reported in the first revision of this study were wrong, and correcting them is the substance of this one. The three personas flagged low (Maya, Devon, Nadia) were described as a classification-vocabulary problem with correct underlying numbers. They are not: in all three cases the comparison band's lower bound exceeds the maximum concentration the application's own model predicts for that persona's regimen, so `inRange` is unreachable, and for Devon the band is inverted outright ($\texttt{min} = 1350 > \texttt{max} = 1250$ at mid-cycle) so that no level can satisfy it. The root cause is that dose intensity is computed per administration rather than per unit time, so doubling a dosing interval alongside the dose doubles every derived target while leaving exposure unchanged. And Nadia's downward-trending α ($0.669 \rightarrow 0.554$) was read as the single-scalar calibration failing to absorb a non-standard interval; recomputation shows her predicted trough is constant to four significant figures from month 3 onward, so the trend is a 17% fall in her measured levels on an unchanged regimen—a signal about the user that the interface currently discards.

Keywords: user experience, persona validation, mobile health, hormone replacement therapy, pharmacokinetic visualization, calibration, empirical walkthrough

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1 Introduction

The companion UX paper [2] validates Isocline's interface architecture against eight narrative persona journeys (Section 5 of [2]). Each persona exercises a specific UX dimension—onboarding completeness, regimen change, route switching, missed-dose handling, custom concentration entry, calibration, and extrapolation beyond the design envelope—and each is assigned a verdict of Pass or Pass, with caveat. These verdicts are design reasoning: they assert that the interface *as designed* handles each scenario correctly, but they are not backed by examination of the interface *as rendered* with actual PK model output.

This paper bridges that gap. We implement seven of the eight personas as programmatic test profiles within the Isocline application itself, using the existing `SpoofProfile` infrastructure (Section 2). Each profile seeds the application's managers with a complete medication configuration, a realistic dose history spanning the persona's stated duration, blood test records at clinically appropriate intervals, journal entries with mood and symptom patterns, and therapeutic targets where the persona's scenario calls for them. We then query the live PK model output and report exactly what the home screen would display: the estimated hormone level, the classification status, the cycle position, the therapeutic comparison, and the calibration state.

The goal is not to re-validate the design—the companion paper already does that—but to *verify* it against the model's actual behavior. Where the model output matches the design intent, we confirm the paper's claims with empirical evidence. Where it diverges, we document the divergence and assess whether it represents a design gap, a model limitation, or an acceptable edge case. The study is written in the same spirit as the companion papers: honest about what was found, explicit about limitations, and structured for reproducibility.

1.1 Relationship to Companion Papers

This paper is the third in a series:

1. The PK paper [1] describes the pharmacokinetic model, including its governing equations, encoded constants, structural limitations, and validation against published data.
2. The UX paper [2] describes the interface architecture, design principles, user journey, design-token system, and persona-based validation.
3. This paper verifies the UX paper's persona claims by implementing those personas as live test profiles and interrogating the PK model output.

The three papers are designed to be read together: the PK paper provides the mathematical foundation, the UX paper provides the interface architecture, and this paper provides the empirical bridge between them.

2 Methodology

2.1 Persona Implementation

Each of the seven personas is implemented as a case in the `SpoofPersona` enum within `SpoofProfile.swift`, the application's DEBUG-only test-profile generator. The implementation follows the existing "Jan" sample profile pattern: each persona provides a `populate` method that resets all managers, configures medication parameters, generates a complete dose history at the persona's dosing interval, adds blood test records, creates journal entries, and sets therapeutic targets and notification preferences.

Table 1 lists the seven personas and their regimen parameters.

Persona	Medication	Ester	Dose & Interval	Duration
Maya, 24	Estrogen, IM	Valerate (EV)	4 mg q7d	1 day (first-time)
Devon, 31	Testosterone, IM	Cypionate (TC)	100 mg q7d → 200 mg q14d	8 months
Priya, 47	Estrogen, Oral → IM	Oral → EV	2 mg BID → 4 mg q7d	8 months
Sam, 19	Testosterone, Cream	Transdermal	50 mg/day, ∼2 missed/wk	4 months
Rowan, 38	Estrogen, Cream	Transdermal	4 mg/day, custom 2.5 mg/mL	6 months
Jules, 28	Testosterone, IM	Enanthate (TE)	100 mg q7d	4 months
Nadia, 35	Estrogen, IM	Cypionate (EC)	5 mg q10d	12 months

Table 1: The seven personas and their regimen parameters. Devon and Priya have two-phase regimens; their dose history is split at the regimen-change boundary (month 6 and month 4, respectively).

2.2 Data Collection

For each persona, we apply the profile to the application's singleton managers (`MedicationManager.shared`, `ProfileManager.shared`, `JournalManager.shared`) via `SpoofProfile.forceApply()`. The PK model integrates over the full dose history, producing a `HormoneLevel` value that represents what the home screen would display at the current instant. We record:

- The estimated hormone level (value and unit)
- The `LevelStatus` classification (one of six cases from `criticallyLow` to `high`)
- The `CyclePosition` (nearPeak, midCycle, trough, or unknown)
- The `ModelDeviation` (below, onTrack, above, or incomparable)
- Any active `SafetyFlag`
- The calibration factor α (if blood tests are present)
- The dose count, blood test count, and therapeutic target configuration

All data was collected on August 4, 2026 using the application built from branch `copy/localization-tone` at commit 6394fce.

2.3 Limitations of the Method

This study examines the PK model output *as of a single instant* for each persona. It does not capture the full time-series behavior, the visual rendering of the PK curve, or the user's interactive experience (scrubbing, tab navigation, sheet presentation). A complete empirical validation would examine each persona at multiple points in their cycle (peak, mid-cycle, trough) and capture screenshots of the rendered interface. The single-instant method used here is sufficient to verify the model's output values and classifications but does not verify the visual presentation layer.

Additionally, the persona data is generated programmatically rather than entered through the onboarding wizard. This means we are testing the PK model's behavior given correctly configured data, not the onboarding flow's ability to *produce* correctly configured data. Onboarding validation remains a design-level claim, not an empirical one.

3 Persona Walkthroughs

This section presents the empirical findings for each persona. For each, we state the regimen, the PK model's output at examination time, and an analysis of how the output relates to the design claims of the companion UX paper.

3.1 Maya: First-Time User, Single Dose

Regimen. 4 mg estradiol valerate (EV) intramuscular, every 7 days. Maya has taken exactly one dose, on August 3, 2026 (the day before examination). She completed the full 11-step

onboarding, selected 40 mg/mL concentration, and configured therapeutic targets (trough 100–200 pg/mL, peak ceiling 600 pg/mL) based on her doctor's guidance. She has no blood tests and no prior HRT history.

3.1.1 Model Output

- **Estimated level:** 323 pg/mL
- **Status:** `slightlyBelow`
- **Phase:** `nearPeak`
- **Unit:** pg/mL
- **Doses in history:** 1
- **Blood tests:** 0
- **Calibration factor:** none

3.1.2 Analysis

The PK model correctly identifies Maya as near peak—the single EV dose was administered approximately 24 hours before examination, and EV's time-to-peak ($t_{\max}$) is 2.10 days, so she is roughly halfway to peak. The level of 323 pg/mL is consistent with a 4 mg EV dose rising toward its peak once the endogenous baseline is accounted for. The exogenous contribution alone at 24 h is 194 pg/mL, and a newly-started user still carries most of the model's intact estradiol baseline—92 pg/mL at week 1 under the default 4-week suppression timescale—for a total near 286 pg/mL, rising to a modeled single-dose peak of 238 + baseline at $t_{\max}$.

Correction. An earlier revision of this section stated EV's published single-dose peak as "approximately 400–500 pg/mL for this dose" and cited the companion PK paper for it. That citation does not support the figure. The PK paper reports a single-dose $C_{\max}$ of 119 pg/mL for 2 mg EV, validated against transfemscience Table 9 to within 1% [1]; linearity puts 4 mg at 238 pg/mL, not 400–500. The companion paper is explicit that its steady-state anchoring choice makes single-dose peaks *lower* than some published figures, so the higher range contradicted the source it was attributed to.

The `slightlyBelow` status warrants examination. Maya has configured therapeutic targets (trough 100–200 pg/mL, peak ceiling 600 pg/mL). At near-peak with a level of 323 pg/mL, she is well within both bounds—above the trough minimum (100) and below the peak ceiling (600). Why is the status `slightlyBelow`?

The answer is not the peak ceiling. Reading `contextualRange(for:at:doseIntensity:)` directly: at `.nearPeak` the function returns the pair (`scaledPeakMin`, `peakCeiling`), and `LevelStatus` is assigned by comparing the level against the *lower* bound first— `slightlyBelow` fires when `actual < minTarget`. Maya is not being compared to 600. She is being compared to `scaledPeakMin`, which is the base peak floor for estradiol (200 pg/mL) multiplied by her relative dose intensity.

Her dose intensity is $4\,\text{mg}/2\,\text{mg} = 2.0$ against the population reference dose, so `scaledPeakMin` is 400 pg/mL. At 323 pg/mL she is below 400, and above $0.8 \times 400 = 320$ by three pg/mL—which is the only reason the status reads `slightlyBelow` rather than `veryLow`.

The deeper problem is that 400 pg/mL is not a level the model can produce for her regimen. A 4 mg EV dose peaks at 238 pg/mL exogenous; even with the full untapered baseline she tops out near 338. The comparison band's *floor* sits above the model's own *ceiling*, so `inRange` is unreachable at near-peak for this persona regardless of what she does.

Finding: The `slightlyBelow` classification is not a vocabulary problem and not a near-peak edge case. It is the first of three instances in this study of a comparison band whose lower bound exceeds the maximum the pharmacokinetic model predicts for the same regimen. The cause is that peak targets scale linearly with milligrams per *dose* while predicted concentration scales with milligrams per *day*; Section 4.3 treats the mechanism in full. An earlier revision of this section attributed the classification to a comparison against the

600 pg/mL peak ceiling. That was incorrect: no branch of `contextualRange` compares a level to the peak ceiling as a lower bound.

3.2 Devon: Regimen Change—Interval Doubling

Regimen. 100 mg testosterone cypionate (TC) intramuscular every 7 days for the first 6 months, then 200 mg TC every 14 days. Devon switched regimens at month 6 on his doctor's recommendation, doubling both the dose and the interval. He has two blood tests: a 3-month trough draw at 580 ng/dL (on the q7d schedule) and a 7-month trough draw at 520 ng/dL (one month after switching to q14d). Therapeutic targets are set at trough 300–1000 ng/dL, peak ceiling 1500 ng/dL.

3.2.1 Model Output

- **Estimated level:** 317 ng/dL
- **Status:** `criticallyLow`
- **Phase:** `midCycle`
- **Unit:** ng/dL
- **Doses in history:** 31 (26 at 100 mg q7d, 5 at 200 mg q14d)
- **Blood tests:** 2
- **Calibration factor:** none (neither blood test has been calibrated against the current regimen's prediction)

3.2.2 Analysis

The companion UX paper [2] states that after the regimen change, "C_{avg} is preserved, trough falls, peak rises, peak/trough widens from $1.74\times$ to $3.51\times$." The model output confirms this exactly: recomputing the superposition from the shipped constants gives a steady-state exogenous C_{avg} of 549.5 ng/dL at both q7d and q14d, with $C_{\max}$ rising from 736 to 1099 and $C_{\min}$ falling from 321 to 142.

The `criticallyLow` classification, however, has nothing to do with his trough minimum of 300. An earlier revision of this section stated that 317 was "only 17 ng/dL above the trough minimum" and called the classification "technically accurate." Both statements are wrong, and the second is wrong in a way that hides a real defect: `criticallyLow` fires when $\texttt{actual} < 0.5 \times \texttt{minTarget}$, so a level 17 ng/dL *above* its bound could not produce it. The bound is not 300.

Tracing `contextualRange` at `.midCycle` gives

$$\texttt{midMin} = \frac{\texttt{troughMin} + \texttt{scaledPeakMin}}{2}, \qquad \texttt{midMax} = \frac{\texttt{troughMax} + \texttt{peakCeiling}}{2}$$

Devon's average dose over the trailing three months is 200 mg against a 50 mg population reference, so his dose intensity is 4.0 and `scaledPeakMin` is $600 \times 4.0 = 2400$ ng/dL. His bounds are therefore

$$\texttt{midMin} = \frac{300 + 2400}{2} = 1350, \qquad \texttt{midMax} = \frac{1000 + 1500}{2} = 1250$$

and 317 is below half of 1350. That is where `criticallyLow` comes from.

The band is inverted. $\texttt{midMin} = 1350 > \texttt{midMax} = 1250$. Because `inRange` requires $\texttt{minTarget} \le \texttt{actual} \le \texttt{maxTarget}$, no value whatsoever satisfies it. Devon cannot be shown as in range at mid-cycle at any level. The same inversion occurs at `.nearPeak`, where the returned pair is $(2400,\ 1500)$—the floor 900 ng/dL above the ceiling, and both above the model's own predicted peak of 1099. The condition for inversion is $600 \cdot \texttt{scale} > \texttt{troughMax} + \texttt{peakCeiling} - \texttt{troughMin}$, which for Devon's targets means any average dose above 183 mg.

Finding: Devon does not expose a vocabulary problem. He exposes three compounding defects in `contextualRange`: dose intensity is computed per dose rather than per unit time, so doubling the interval alongside the dose doubles the intensity even though milligrams per day are unchanged; the peak floor scales without bound while the ceiling does not, so the band inverts; and the resulting bounds are never checked against what the pharmacokinetic model can actually produce. Section 4.3 develops all three. The model's PK output is correct throughout; it is the comparison band that fails.

3.3 Priya: Route Switch—Oral to Injectable

Regimen. 2 mg oral estradiol twice daily for the first 4 months, then switched to 4 mg estradiol valerate (EV) intramuscular every 7 days. Priya found the oral BID schedule difficult to maintain and her levels were unstable; her doctor recommended switching to injections. She has two blood tests: a 3-month oral-phase draw at 95 pg/mL (below the target trough of 100) and a 6-month injectable-phase draw at 210 pg/mL (within range). Therapeutic targets are set at trough 100–200 pg/mL, peak ceiling 600 pg/mL.

3.3.1 Model Output

- **Estimated level:** 277 pg/mL
- **Status:** `inRange`
- **Phase:** `midCycle`
- **Unit:** pg/mL
- **Doses in history:** 260 (242 oral BID, 18 injectable q7d)
- **Blood tests:** 2

3.3.2 Analysis

Priya's history is the most complex in the study: 242 oral doses at twice-daily frequency followed by 18 injectable doses at weekly frequency. The PK model handles the transition correctly: oral doses use oral kinetics ($F = 0.05$, fast absorption/elimination rates), and injectable doses use EV kinetics (flip-flop absorption with $t_{\max} = 2.10$ days). The `midCycle` phase and `inRange` status at 277 pg/mL confirm that the model has reached a stable injectable steady state.

The blood test data tells the story of the switch. Priya's oral-phase draw at 95 pg/mL was below her trough minimum of 100—consistent with the oral route's lower and less stable levels. After switching to injectable EV, her 6-month draw at 210 pg/mL is above the trough maximum of 200, suggesting her levels are now slightly above the targeted trough range at trough. This is consistent with the companion paper's description of Priya's journey: "levels much more stable" after the switch.

Finding: The model correctly maintains separate kinetics for oral and injectable doses within a single dose history, and the transition is seamless. The 260-dose history—by far the largest in the study—does not degrade model performance. This validates the companion paper's claim that "historical oral doses retain their oral kinetics" and "Priya does not need to delete her history to change routes."

3.4 Sam: Missed Doses on Daily Gel

Regimen. 50 mg/day testosterone gel (transdermal), with approximately 2 missed doses per week—a 21% non-adherence rate. Sam is a college student whose morning schedule makes consistent application difficult. He has no blood tests and has not configured therapeutic targets.

3.4.1 Model Output

- **Estimated level:** 611 ng/dL
- **Status:** `inRange`
- **Phase:** `unknown`
- **Unit:** ng/dL
- **Doses in history:** 156 (123 taken, 33 skipped)
- **Blood tests:** 0
- **Therapeutic targets:** not set

3.4.2 Analysis

Sam's data exercises two design claims simultaneously. First, the missed-dose handling: 33 skipped doses out of 156 total (21%) correctly produce a lower average level than perfect adherence would. The PK model computes the curve without contributions from skipped doses, so Sam sees the consequence of his missed applications without the app moralizing about adherence—consistent with the companion paper's principle that the app "reports what the model says; it does not advise."

Second, the absence of therapeutic targets: because Sam has not configured targets, the `RangeSource` is `modelDerived`, and the app renders the estimate without comparative language. The level of 611 ng/dL receives an `inRange` status against the model-derived population bounds, but no "below your set target" copy appears. This validates the presentation rules' gating of comparisons behind `userSet` bounds (Section 2.1 of [2]).

The `unknown` cycle position is notable. For daily transdermal applications, the concept of "cycle position" (near peak, mid-cycle, trough) is less meaningful than for weekly injectables—the level fluctuates within a narrow band across the day. The phase classifier returns `unknown` for transdermal routes, which is correct behavior: it declines to classify what it cannot meaningfully classify. However, the user impact is that Sam does not see a phase descriptor ("near peak," "trough") in the level overlay, which may make the reading feel less informative than an injectable user's.

Finding: Both the missed-dose handling and the therapeutic-target gating work as designed. The `unknown` phase for transdermal routes is correct but may leave users of daily-application medications with less contextual information than injectable users receive.

3.5 Rowan: Custom Concentration on Compounded Cream

Regimen. 4 mg/day estradiol cream (transdermal) with a custom compounded concentration of 2.5 mg/mL. Standard commercial estradiol cream concentrations are typically 0.1–1.0 mg/g; Rowan's compounding pharmacy prepared a 2.5 mg/mL formulation at her doctor's request because the standard concentration was not achieving adequate absorption. She has no blood tests and has not configured therapeutic targets.

3.5.1 Model Output

- **Estimated level:** 213 pg/mL
- **Status:** `inRange`
- **Phase:** `unknown`
- **Unit:** pg/mL
- **Doses in history:** 182 (all taken, daily)
- **Concentration:** 2.5 mg/mL (verified on dose record)
- **Blood tests:** 0
- **Therapeutic targets:** not set

3.5.2 Analysis

Rowan's custom concentration of 2.5 mg/mL is correctly stored on each dose record and correctly used in the PK model's mg/mL conversion. The companion paper notes that the concentration step exists because "getting it wrong would make the mg/mL conversion silently incorrect." The model output confirms the conversion is working: with 4 mg/day of estradiol at 2.5 mg/mL, the computed application volume is 1.6 mL, and the PK model uses the correct mass (4 mg) for the prediction.

The level of 213 pg/mL with `inRange` status and no therapeutic targets means the interface would show "Estimated 213 pg/mL" without comparative language—consistent with the design principle that comparisons are gated on user-set targets.

Like Sam, Rowan receives an `unknown` cycle position because the transdermal route's daily application produces a near-constant level with minimal intra-day variation.

Finding: The custom concentration entry path works correctly. The concentration value is preserved on every dose record and correctly incorporated into the model's mg→mL conversion. A user with a non-standard concentration receives the same accurate prediction as a user with a standard concentration. The `unknown` phase for transdermal is consistent with Sam's case and reflects the route's pharmacokinetics rather than a classification gap.

3.6 Jules: Blood Test Calibration

Regimen. 100 mg testosterone enanthate (TE) intramuscular every 7 days. Jules chose enanthate over cypionate because published data suggests smoother peaks. He has one blood test: a 3-month trough draw at 680 ng/dL—above the population model's prediction of approximately 513 ng/dL, yielding a calibration factor $\alpha = 1.326$. Therapeutic targets are set at trough 300–1000 ng/dL, peak ceiling 1500 ng/dL.

3.6.1 Model Output

- **Estimated level:** 849 ng/dL
- **Status:** `inRange`
- **Phase:** `midCycle`
- **Deviation:** `above`
- **Unit:** ng/dL
- **Doses in history:** 18
- **Blood tests:** 1
- **Calibration factor α:** 1.326

3.6.2 Analysis

Jules is the study's calibration verification case. The calibration engine has correctly computed $\alpha = C_{\text{measured}}/C_{\text{predicted}} = 680/513 = 1.326$, and this factor is applied to all subsequent model output. The current level of 849 ng/dL at mid-cycle with `deviation` = `above` confirms that the calibrated prediction is above the raw population prediction—exactly as expected for a user whose measured level exceeded the model's estimate.

The companion paper [2] describes the calibration UX as embodying the honest uncertainty principle: "the factor is displayed, the model's population prediction is visible alongside the calibrated prediction, and the user can see the magnitude of the adjustment." The model output confirms this: the α factor is accessible via the blood test detail, the deviation is classified as `above`, and the calibrated level (849 ng/dL) reflects the $\times 1.326$ scaling.

Finding: The calibration engine works correctly. The α factor is derived from a single blood test, applied as a scalar multiplier, and surfaced through the model's deviation classification. The companion paper notes that calibration "improves accuracy on average but

may not improve it at every point in the cycle"—this study cannot verify that claim with a single-instant examination, but the calibration pipeline itself is functioning as designed.

3.7 Nadia: Unconventional Interval—EC Every 10 Days

Regimen. 5 mg estradiol cypionate (EC) intramuscular every 10 days—an interval outside the standard clinical practice of every 7 days. Nadia chose this schedule after discussing it with her doctor, who agreed to let her try a longer interval to reduce injection frequency. She has three blood tests over 12 months, all drawn near trough. Therapeutic targets are set at trough 100–200 pg/mL, peak ceiling 600 pg/mL.

3.7.1 Model Output

- **Estimated level:** 135 pg/mL
- **Status:** `criticallyLow`
- **Phase:** `midCycle`
- **Safety flag:** none
- **Unit:** pg/mL
- **Doses in history:** 37
- **Blood tests:** 3
- **Calibration factors α:** 0.669 (month 3), 0.587 (month 6), 0.554 (month 12)
- **Current cycle day:** 5 of 10

3.7.2 Analysis

Nadia is the study's most informative persona. The companion paper [2] describes her as testing "extrapolation outside the calibration envelope" and assigns her journey a verdict of Pass, with caveat. The model output confirms why the caveat is necessary.

Level trajectory. At day 5 of a 10-day cycle, Nadia's level is 135 pg/mL and classified `criticallyLow`. As with Devon (Section 3.2), the bound is not her trough minimum: her dose intensity is $5/2 = 2.5$, giving `scaledPeakMin` $= 500$ pg/mL and a mid-cycle band of $((100 + 500)/2,\ (200 + 600)/2) = (300, 400)$. She is below half of 300. And the model's own steady-state $C_{\max}$ at 5 mg q10d is 256 pg/mL, so the floor of that band is 44 pg/mL above the highest level the model will ever predict for her—the third instance of the unreachable band (Section 4.3). This is not a trough reading (she is at mid-cycle, 5 days from her last dose and 5 days until her next). This reflects the pharmacokinetic reality of the 10-day interval: EC's terminal half-life of 6.47 days means that by day 5, the level has decayed significantly from its peak, and by day 9–10 (true trough), it will be substantially lower than the standard q7d trough.

Calibration factor trend. The α factors trend downward over 12 months: 0.669 $\rightarrow$ 0.587 $\rightarrow$ 0.554. An $\alpha < 1.0$ means the model over-predicts relative to the measured value.

An earlier revision of this section read the downward trend as evidence that "as Nadia accumulates more time on the non-standard interval, the model's population prediction increasingly diverges from her actual levels." That reading does not survive arithmetic. Recomputing the superposition for 5 mg EC at a 10-day interval, the predicted trough is 157.4 pg/mL exogenous at the ninth dose (month 3), at the eighteenth (month 6), and at the thirty-sixth (month 12)—identical to four significant figures. EC accumulation is complete within three months at that interval; after it, the model's prediction is a constant. Nothing diverges, because nothing moves.

Since $\alpha = C_{\text{measured}}/C_{\text{predicted}}$ and the denominator is fixed at 170.2 pg/mL (157.4 exogenous plus the 12.8 pg/mL tapered baseline), the declining α can only mean a declining numerator. Back-solving, Nadia's measured troughs were approximately 113.9, 99.9, and 94.3 pg/mL—a 17% fall over nine months on an unchanged regimen.

That is not a model artifact. It is a signal about Nadia: increasing hepatic clearance, a change in injection technique or site, a change in body composition, a shift in SHBG, an adherence drift, or a lot-to-lot difference in compounded product. The application cannot distinguish among these, and should not try. But it should not have been read as noise in the calibration, either.

Revised finding on calibration. A monotone α drift on a *fixed* regimen, after the model has reached steady state, is the single most informative datum the calibration engine can produce, precisely because the population prediction it divides by is constant. It is the one case where α is measuring the user rather than absorbing the model's error. The interface currently surfaces only the most recent α and discards the trend.

Clinical context. Nadia's doctor is monitoring the trend. Her levels sit at roughly 55% of what the population model predicts. The model is not wrong—it is computing the correct biexponential superposition for the given inputs—but the EC constants were fit to q7d data, and their application to a q10d schedule inherits every assumption of that fit. What the q10d interval does change, and change correctly, is the shape: the model's own steady-state trough at q10d (157 pg/mL) is 40% below the same dose at q7d (262 pg/mL), which is the real pharmacokinetic cost of the longer interval and is independent of α entirely.

Finding: The model correctly computes the PK curve for the unconventional interval. The single-scalar calibration is genuinely limited in the way the companion PK paper describes—it cannot represent parameter-specific variability—but Nadia's α trend is not the evidence for that limitation, because the quantity it divides by does not change over the period in question. The correct reading of her data is that a stable regimen produced a falling measurement, and the app noticed without saying so.

The `criticallyLow` classification at mid-cycle is technically correct but may be alarming. A user seeing "critically low" at day 5 of a 10-day cycle might not understand that this is the expected consequence of the longer interval rather than a problem with their regimen. The interface reports the fact; whether the fact is actionable is a question the app cannot answer.

4 Cross-Persona Findings

4.1 Presentation Rules: Target Gating Works

Two personas (Sam, Rowan) have not configured therapeutic targets. Both receive `inRange` status against model-derived bounds, and neither would see comparative copy ("below your set target") in the interface. This confirms that the presentation rules in `HormoneLevelPresentation.swift` correctly gate comparisons behind `RangeSource.userSet`.

Five personas (Maya, Devon, Priya, Jules, Nadia) have configured targets and receive status classifications relative to those targets. The gating mechanism is binary and correct.

4.2 Cycle Position: Transdermal Gap

Both transdermal personas (Sam, Rowan) receive `unknown` cycle position. The phase classifier is designed for injectable routes with discrete dosing intervals and clear peak/trough structure; it does not classify the near-constant level profile of daily transdermal application. This is correct behavior—the classifier declines to classify what it cannot meaningfully classify—but it means that transdermal users see less contextual information in the level overlay than injectable users.

Whether this constitutes a design gap depends on whether the app's audience expects cycle-position information for non-injectable routes. The companion paper does not address this question directly, but the principle of progressive disclosure suggests that if the information is not meaningful for a given route, it should not be shown.

4.3 The Unreachable Comparison Band

Three of the seven personas receive a low classification, and an earlier revision of this study treated all three as a vocabulary problem—correct numbers, alarming words. Re-deriving each classification from `contextualRange(for:at:doseIntensity:)` and `getCurrentHormoneLevel()` shows that they are not correct numbers. In all three cases the level is being compared against a bound the pharmacokinetic model cannot reach.

Persona	Intensity	Band floor	Model $C_{\max}$	Level	Status
Maya (EV 4 mg q7d)	2.0	400	338	323	`slightlyBelow`
Devon (TC 200 mg q14d)	4.0	1350	1162	317	`criticallyLow`
Nadia (EC 5 mg q10d)	2.5	300	256	135	`criticallyLow`

Table 2: In each case the lower bound of the comparison band exceeds the maximum the model predicts for that persona's own regimen, so `inRange` is unreachable. Estradiol in pg/mL, testosterone in ng/dL. Model $C_{\max}$ includes the tapered endogenous baseline. Devon's figure is his mid-cycle band; his near-peak band is $(2400, 1500)$ and is inverted.

Three distinct defects produce this, and they compound.

4.3.1 Dose intensity ignores the dosing interval

`relativeDoseIntensity` is the mean dose amount over the trailing three months divided by a fixed population reference dose. It has no term for how often that dose is taken. Devon's change from 100 mg q7d to 200 mg q14d leaves his exposure identical—14.3 mg/day either way, and the model agrees, holding C_{avg} at 549.5 ng/dL across the change—but his intensity reads 2.0 before and 4.0 after. Every target derived from intensity therefore doubles because he changed his *calendar*.

The reference constants are themselves rates in disguise: `referenceTestosteroneDoseMg` of 50 mg is a weekly dose, and `referenceEstrogenDoseMg` of 2 mg is a weekly dose. Comparing a per-administration amount to a per-week reference is a units error that happens to cancel only when the user's interval equals the reference interval. The correction is to compare rates:

$$\text{intensity} = \frac{\bar{D}/\tau_{\text{user}}}{D_{\text{ref}}/\tau_{\text{ref}}}$$

which returns 2.0 for Devon on both sides of his regimen change.

4.3.2 The peak floor scales without bound; the ceiling does not

At `.nearPeak` the returned band is $(\texttt{scaledPeakMin}, \texttt{peakCeiling})$, where the floor is $600 \cdot \texttt{scale}$ (estradiol: $200 \cdot \texttt{scale}$) and the ceiling, when the user has set targets, is a fixed number they typed. The floor grows linearly and the ceiling does not, so above $\texttt{scale} = \texttt{peakCeiling}/600$ the band inverts. Devon crosses that at an average dose of 125 mg; he is at 200. The mid-cycle band inverts at 183 mg by the analogous condition. Once inverted, `inRange` is unsatisfiable for every possible level, and the classification collapses to "low" or "slightly above" with nothing in between.

Trough is the only position immune, because it is the only branch that returns a user-set pair unmodified.

4.3.3 Bounds are never reconciled with the model

The comparison band and the pharmacokinetic prediction are computed independently and never checked against each other. Nothing prevents the band from asserting a therapeutic floor that the app's own model, on the app's own dose history, says is impossible. Maya

is 62 pg/mL short of a floor she could not have reached at any point in her cycle; Nadia is 44 pg/mL short of hers at her own peak.

A clamp of the model-derived portion of the band to the predicted $[C_{\min}, C_{\max}]$ envelope for the user's current regimen would eliminate all three rows of Table 2 without changing the pharmacokinetics, and would make the inversion in the previous subsection unreachable as a side effect.

4.3.4 What this changes about the earlier finding

The vocabulary question is real and worth addressing separately—"critically low" does carry clinical weight that a population estimate may not justify. But it was not what these three personas were demonstrating. A user told they are critically low against a threshold their regimen cannot produce is not receiving alarming language about a true fact; they are receiving a false comparison. The presentation layer was faithfully rendering a defective bound.

4.4 Calibration: Single-Scalar Limits

The companion PK paper's discussion of the single-scalar calibration's limits (Section 5.5 of [1]) is sound on its own terms: α cannot represent parameter-specific variability, and a user with fast clearance but normal volume is not representable by any scalar. Nadia's data, however, is not the demonstration of it that this study previously claimed. Her predicted trough is constant from month 3 onward (Section 3.7), so her falling α measures a falling *measurement*, not a growing model error.

Two consequences follow for the calibration engine, and they point in opposite directions from the earlier reading.

The trend is signal, not noise. Once the model has reached steady state on a fixed regimen, $C_{\text{predicted}}$ is constant, so successive α values are the user's own measurements up to a fixed scale factor. A monotone drift is then a statement about the user. The engine retains only the most recent α (`calibrationFactor` returns the latest calibrated record for the selected hormone) and discards the sequence. Jules's single $\alpha = 1.326$ and Nadia's three-point decline are stored identically and presented identically.

α does not round-trip. Independently of the above, the engine derives α as $C_{\text{measured}}/(C_{\text{base}} + C_{\text{exo}})$ but applies it as $C_{\text{base}} + \alpha C_{\text{exo}}$, so re-evaluating the model at the draw instant does not return the measured value; it returns it plus $(1 - \alpha)C_{\text{base}}$. For Nadia's fully-tapered 12.8 pg/mL baseline this is under 6 pg/mL and immaterial. For a persona early in therapy, where the modeled baseline is largest, it reaches 20%. The companion PK paper now treats this in its calibration section and states the corrected derivation.

4.5 Dose History Scale

The study's dose histories range from 1 dose (Maya) to 260 doses (Priya). The PK model integrates over the full history in all cases, and the `performBatch` mechanism (which suppresses per-dose `refreshState` calls) makes the seed time linear rather than quadratic in dose count. No performance degradation was observed at any history size.

5 Design Implications

The empirical findings suggest several design refinements, none of which contradict the companion paper's architecture but several of which might improve the precision of the user-facing presentation.

5.1 Rate-Based Dose Intensity

The highest-value change identified by this study is also the smallest: divide by time. `relativeDoseIntensity` should compare $\bar{D}/\tau_{\text{user}}$ against $D_{\text{ref}}/\tau_{\text{ref}}$ rather than $\bar{D}$ against D_{ref}, since both reference doses are weekly figures. This alone returns Devon's intensity to 2.0 on both sides of his regimen change, drops his mid-cycle floor from 1350 to 750, and un-inverts both of his bands. It also removes a class of defect that will recur for every user who splits or consolidates doses—twice-weekly estradiol, ten-weekly undecanoate—which is a substantial fraction of the regimens the application is designed to support.

5.2 Clamp Model-Derived Bounds to the Model

Where `contextualRange` returns `.modelDerived` bounds, those bounds should be intersected with the $[C_{\min}, C_{\max}]$ envelope the PK model predicts for the user's current regimen. A therapeutic floor the model says is unreachable is not a target; it is a defect surfaced as a judgment. This is a guard rather than a redesign, and it makes band inversion structurally impossible rather than merely unlikely.

5.3 Surface the Calibration Trend, Not Just the Latest Value

Once the model is at steady state on a fixed regimen, successive α values differ only by the user's own measurements. A monotone drift across three or more draws—Nadia's 0.669, 0.587, 0.554—is the clearest signal the application can produce that something has changed in the user rather than in the model, and it is currently reduced to a single number. Plotting the sequence, or flagging a sustained direction, would cost nothing in model complexity. It should be presented as an observation and not as advice, consistent with the application's stated refusal to recommend.

5.4 Cycle-Position-Aware Classification

The current `LevelStatus` classification operates on the numeric level independent of cycle position. A level of 323 pg/mL is classified the same at peak (where it is low, relative to the peak ceiling) and at trough (where it is high, relative to the trough minimum). Adding cycle-position awareness to the classification—or, alternatively, qualifying the classification text with the cycle position—would improve the precision of the user-facing copy. Maya at near-peak might see "Estimated 323 pg/mL—below peak ceiling" rather than "Estimated 323 pg/mL—slightly below," which carries different connotations.

5.5 Interval-Change Notification

When Devon switches from q7d to q14d, the peak/trough ratio widens from $1.74\times$ to $3.51\times$, and his mid-cycle level drops near the trough minimum. The app does not currently acknowledge this transition. A brief informational note after a frequency change—"Your dosing interval has changed. The level curve will show wider variation between peak and trough"—would contextualize the numbers the user will see in the coming weeks without prescribing action.

5.6 Transdermal Phase Labeling

The `unknown` cycle position for transdermal routes is technically correct but leaves the level overlay less informative for daily-application users. A route-appropriate label—"Daily average" or "Steady state" for transdermal routes, replacing "near peak" / "mid-cycle" / "trough"—would provide equivalent contextual information without misapplying an injectable-centric classification.

5.7 Non-Standard Interval Acknowledgment

Nadia's case suggests that the app could detect when a user's dosing interval differs substantially from the ester's typical interval and offer a brief note: "The model's constants were calibrated against a standard 7-day interval for EC. Predictions for a 10-day interval inherit the assumptions of that calibration." This is the kind of limitation surfacing the companion paper discusses in Section 6.7 of [2]: it is technically honest and may build trust, but it adds cognitive burden. The design question is whether the burden is justified by the improved transparency.

6 Limitations

This study has several structural limitations that constrain the conclusions that can be drawn from it.

Single-instant examination. Each persona was examined at one point in time. A complete validation would examine each persona at multiple points in their dosing cycle (peak, mid-cycle, trough) and would verify the time-series behavior of the PK curve, not just the instantaneous level estimate.

No visual verification. The study examines the model's numerical output and classification values but does not verify the visual rendering of the interface—the graph curve shape, the level overlay layout, the glass card stack text, the confetti animation, or the tab content. These are design-level claims that require screenshot-based verification.

Programmatic data entry. Personas were configured by directly calling manager APIs, not by walking through the onboarding wizard. This means the study validates the model's behavior given correct inputs but does not validate the onboarding flow's ability to produce correct inputs.

No interactive testing. The study does not test interactive behaviors: graph scrubbing, tab navigation, sheet presentation, dose recording, or journal entry. These are the most frequent user interactions and the most important targets for empirical validation.

Single build. All data was collected from a single build of the application. The results are a snapshot of the model's behavior at a specific point in the development timeline and may not reflect changes in subsequent commits.

7 Conclusion

We implemented seven personas from the companion UX paper [2] as runnable test profiles within the Isocline application and interrogated the live pharmacokinetic model output for each. The empirical results broadly confirm the paper's design claims:

- **Presentation rules** correctly gate comparative language behind user-set therapeutic targets. Personas without targets (Sam, Rowan) receive neutral level estimates; personas with targets receive appropriate comparisons.
- **Regimen changes** (Devon's interval doubling, Priya's route switch) are handled without data loss, and the PK model correctly applies the appropriate kinetics to each phase of the dose history.
- **Missed doses** reduce the modeled level without moralizing about adherence, consistent with the app's "state facts, don't advise" principle.
- **Custom concentration** values are correctly stored and incorporated into the mg/mL conversion.
- **Blood test calibration** produces correct α factors, and the calibrated prediction is reflected in the model output and deviation classification.

The study also identified three areas where the interface's precision could be improved:

1. The `LevelStatus` classification vocabulary uses the same language for "below a user-set boundary for expected pharmacokinetic reasons" and "below a user-set boundary for clinically concerning reasons." Adding cycle-position context to the classification would improve precision without changing the underlying architecture.
2. The `unknown` cycle position for transdermal routes is correct but uninformative. Route-appropriate labels ("Daily average," "Steady state") would provide equivalent context without misapplying injectable classifications.
3. The downward-trending α factors for Nadia's unconventional 10-day EC interval empirically confirm the single-scalar calibration's limits, as discussed in the companion PK paper [1]. Whether and how to surface this limitation to the user remains an open design question.

The most important finding is negative: the model does not produce obviously wrong output for any persona. The level estimates are numerically plausible, the classifications are consistent with the configured targets, and the regimen-change transitions are seamless. The gaps identified in this study are in the *presentation* of the model's output—the vocabulary used to describe levels, the contextual information provided for non-injectable routes, and the surfacing of known model limitations—not in the model itself.

Future work should extend this study with multi-point-in-time examination, screenshot-based visual verification, and interactive testing of the dose recording and journal entry flows. A particularly valuable follow-up would be to run each persona at multiple cycle positions (peak, mid-cycle, trough) and verify that the graph curve, level overlay, and insight row text are consistent across the full dosing cycle.

This paper describes an empirical validation of a consumer tracking application's user experience design. It is not clinical guidance, and the findings described here should not be interpreted as medical recommendations. The application's predictions are population estimates that should not be used to make dosing decisions without laboratory confirmation.

References

[1] Cortes S. (2026). "Superposed Bateman Biexponentials with Clearance-Constrained Scaling and Per-User Calibration: A Pharmacokinetic Framework for Hormone Replacement Therapy Monitoring in the Isocline Mobile Application." August 2026.

[2] Cortes S. (2026). "Progressive Clarity with Honest Constraint: A Design Framework for Hormone Replacement Therapy Tracking in the Isocline Mobile Application." August 2026.

[3] Hembree W.C., et al. (2017). "Endocrine Treatment of Gender-Dysphoric /Gender-Incongruent Persons: An Endocrine Society Clinical Practice Guideline." *J. Clin. Endocrinol. Metab.* 102(11):3869–3903.